Break-even growth ladder for structured-base pair-sum coincidences

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Track: generic decomposition (bounded by √r; cannot beat rho). This measures a relation-yield statistic, not a cost: no ranking against k-sum, no priced IC cost, no dashboard point (PROTOCOL.md §1).

Result. `SUB_BREAK_EVEN_AT_SCALE`. On prime-order curves of 16 to 32 bits, the pair-sum coincidence excess κ of x-coordinate-structured bases does not grow with N.

- **Slopes.** d ln κ / d ln N has a 95% interval inside [–0.006, +0.008] for all six geometry-θ combinations (window, rank-2 and rank-3 label GAP, at θ = 0.30 and 0.35). The largest upper bound of a 95% interval is **0.008**, against the break-even exponents δ*(h) = 1/3, 1/5, 1/7: 41×, 25× and 18× below them.
- **Level.** Every rung's κ has its 95% interval inside [0.92, 1.08] (point estimates 0.967 to 1.034).
- **Controls.** All gates pass. The Z/N interval recovers slopes 0.699 and 0.649 (exact 0.70 and 0.65), and the weighted plants recover their integer-count slopes to within 0.001.
- **Reading.** Registered as `exhausted_at_scale` at 32 bits (`p < 2^32`). It is not a proof, and it says nothing about growth beyond the rungs measured.

Distance remaining. A growing harvest wins only if κ > 1.27·c·N^(1/(2h–1)). At 32 bits that is 1,874·c (h = 2), 101·c (h = 3) and 29·c (h = 4), against a measured κ of at most 1.07 (T3 below): 27× to 1,745× short for c = 1. The measured κ is flat, so the gap widens with N.

Public synthetic curves; no security claim. Registered in `PROTOCOL.md` before any pilot or main run (hashes in `REGISTRATION.sha256`).

Labels. `[measured]`: native `coincidence_ladder` outputs in this directory. `[derived]`: arithmetic on those outputs. `[recorded]`: values read from the EXP-PFDR-ca5e01 record.

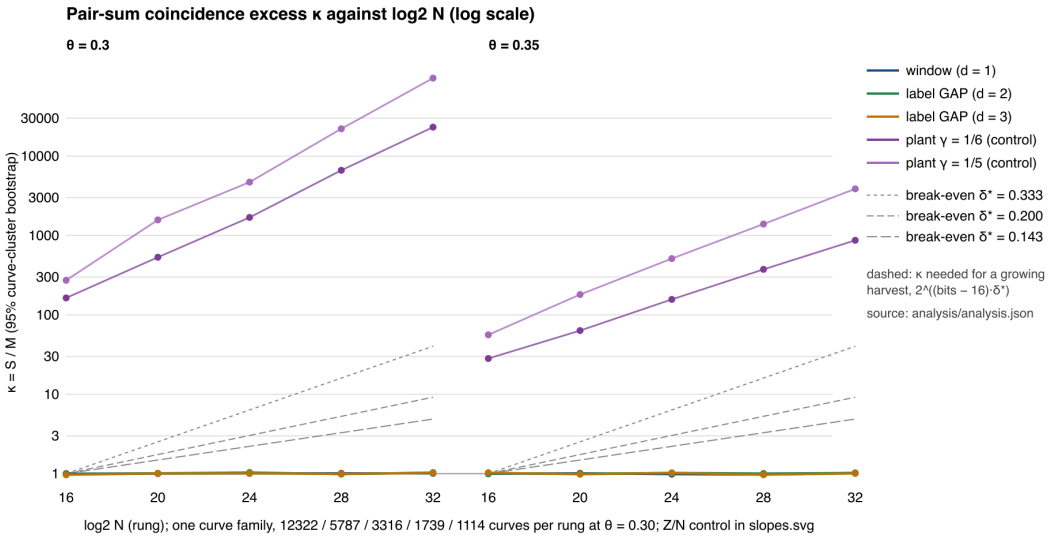


Figure 1: κ per rung for the three real geometries (flat at 1) and the two planted controls (growing), at both θ, with 95% curve-cluster bootstrap bars. Dashed: the κ a harvest would need, 2^((bits − 16)·δ*). Source: kappa_ladder.svg.

1. What was measured

- **Statistic.** For a symmetric base F of exactly b points, C(F) is the number of pairs of unordered pairs {P, Q} ≠ {R, S} with P + Q = R + S (P ≠ Q, P ≠ −Q), counted exactly: a 4-term relation among base points. κ = S/M is the structured count over the mean of 20 matched random symmetric bases.
- **Bases.** Window (consecutive x), rank-2 and rank-3 label GAP, as in EXP-PFDR-ca5e01. All three were run; none was selected afterwards. b = 2·round(2^(bits·θ)/2) with θ ∈ {0.30, 0.35}.
- **Ladder.** 16, 20, 24, 28, 32 bits on prime-order curves with random (a, b); the order is proved by a unique prime multiple in the Hasse interval. The pilot sized each cell for a 2% relative standard error (T4); no cap bound, and the achieved precision matches the prediction.
- **Slope.** OLS of ln κ on the rung's mean ln N, with a joint curve-cluster bootstrap (2000 resamples).

2. Result `[measured]`

Real geometries (T1)

θ = 0.3

geometry	κ at 16 bits	κ at 20 bits	κ at 24 bits	κ at 28 bits	κ at 32 bits	slope [95%]	class
window (d = 1)	1.001 [0.965, 1.040]	1.003 [0.965, 1.044]	1.007 [0.968, 1.046]	1.010 [0.969, 1.051]	1.008 [0.970, 1.047]	0.001 [–0.004, 0.005]	null
label GAP (d = 2)	0.986 [0.949, 1.024]	1.012 [0.971, 1.050]	1.034 [0.993, 1.076]	0.987 [0.945, 1.028]	1.033 [0.991, 1.074]	0.002 [–0.002, 0.007]	null
label GAP (d = 3)	0.967 [0.930, 1.002]	1.006 [0.966, 1.047]	1.011 [0.973, 1.048]	0.992 [0.950, 1.034]	1.023 [0.986, 1.063]	0.004 [–0.001, 0.008]	null

θ = 0.35

geometry	κ at 16 bits	κ at 20 bits	κ at 24 bits	κ at 28 bits	κ at 32 bits	slope [95%]	class
window (d = 1)	0.997 [0.956, 1.037]	1.011 [0.973, 1.050]	0.986 [0.944, 1.026]	0.973 [0.938, 1.008]	1.020 [1.002, 1.038]	0.000 [–0.004, 0.004]	null
label GAP (d = 2)	1.010 [0.968, 1.050]	0.996 [0.956, 1.037]	1.022 [0.983, 1.062]	1.004 [0.962, 1.044]	1.020 [1.003, 1.038]	0.001 [–0.003, 0.005]	null
label GAP (d = 3)	1.029 [0.990, 1.069]	0.982 [0.946, 1.018]	1.020 [0.981, 1.057]	0.971 [0.929, 1.014]	1.006 [0.986, 1.026]	–0.002 [–0.006, 0.002]	null

All six slope intervals contain 0, and the largest upper bound is 0.008. The registered predictions P1 and P2 hold (PROTOCOL.md §9): P1 is |slope| ≤ 0.05, upper < 0.07, at least five of six intervals containing 0, and every κ point estimate in [0.95, 1.05]. They were written after a dry run on other curves, so they are not independent predictions (PROTOCOL.md §11).

Controls (T2)

θ	control	slope [95%]	expected	κ at 16 bits	κ at 32 bits
0.3	Z/N interval	0.699 [0.698, 0.700]	$1 - \theta = 0.70$	1141.3	2666639.7
0.3	plant $\gamma = 0$ (random)	–0.001 [–0.005, 0.004]	0.000 (integer count)	1.02	1.00
0.3	plant $\gamma = 1/6$	0.448 [0.447, 0.450]	0.448 (integer count)	163.54	23175.80
0.3	plant $\gamma = 1/5$	0.518 [0.517, 0.519]	0.518 (integer count)	272.84	96085.28
0.3	null vs null	0.000 [–0.004, 0.004]	0	0.997	0.997
0.35	Z/N interval	0.649 [0.647, 0.651]	$1 - \theta = 0.65$	660.2	893378.8
0.35	plant $\gamma = 0$ (random)	0.001 [–0.002, 0.005]	0.000 (integer count)	1.00	1.01
0.35	plant $\gamma = 1/6$	0.311 [0.306, 0.316]	0.310 (integer count)	28.23	870.68
0.35	plant $\gamma = 1/5$	0.378 [0.373, 0.383]	0.378 (integer count)	56.13	3872.38
0.35	null vs null	–0.003 [–0.006, 0.001]	0	1.003	0.981

Z/N is the known positive, with slope $1 - \theta$. The plants' reference slope comes from an integer count of the rows' coincidences with no group arithmetic, so the match also checks the group-sum kernel. A random $\gamma = 0$ plant and null-vs-null both read slope 0 (gate G1: 0 of 4 intervals exclude 0).

Distance remaining (T3) `[derived]`

bits	mean N	1.27·N^(1/3) (h = 2)	1.27·N^(1/5) (h = 3)	1.27·N^(1/7) (h = 4)	largest real κ upper bound
16	4.89e+4	46	11	6	1.069
20	7.82e+5	117	19	9	1.050
24	1.26e+7	295	33	13	1.076
28	2.01e+8	744	58	20	1.051
32	3.21e+9	1874	101	29	1.074

Cells and precision (T4)

bits	θ	b	curves	pilot M1	pilot V1	pre-dicted rel. sd	achieved rel. sd of S (win-dow)
16	0.3	28	12322	1.18	6.56	0.020	0.019
16	0.35	48	1176	12.06	65.16	0.020	0.020
20	0.3	64	5787	2.68	15.83	0.020	0.021
20	0.35	128	362	43.01	254.77	0.020	0.020
24	0.3	148	3316	4.82	29.35	0.020	0.020
24	0.35	338	119	134.63	820.74	0.020	0.020
28	0.3	338	1739	8.42	46.99	0.020	0.021
28	0.35	892	42	404.93	2587.59	0.020	0.020
32	0.3	776	1114	14.14	84.87	0.020	0.020
32	0.35	2352	40	1211.37	6792.57	0.011	0.011

Stage 1: what was already recorded (T6) `[recorded]`

The recorded ca5e01 cells (16 to 22 bits; 3 or 4 cells per group; relative sd about 4.5%) give slopes consistent with 0, but only to about ±0.05. Only θ = 0.30 has three powered rungs in any group.

θ	d	cells	slope [95%]
0.3	1	4	0.016 [–0.013, 0.046]
0.3	2	3	0.018 [–0.025, 0.063]
0.3	3	3	–0.040 [–0.088, 0.007]

The new ladder reaches 32 bits with 2% cells and tightens the bound from upper bounds of 0.046 and 0.063 (θ = 0.30, d = 1 and 2) to 0.008.

Overlap with ca5e01 (T7) `[measured]`

bits	d	ca5e01 κ [95%]	this study κ [95%]	intervals overlap
16	1	0.966 [0.879, 1.056]	1.001 [0.965, 1.040]	yes
16	2	1.002 [0.915, 1.086]	0.986 [0.949, 1.024]	yes
20	1	1.023 [0.935, 1.121]	1.003 [0.965, 1.044]	yes
20	2	1.053 [0.962, 1.146]	1.012 [0.971, 1.050]	yes
20	3	1.005 [0.919, 1.096]	1.006 [0.966, 1.047]	yes

The curve generators differ, so this is statistical agreement, not a replication. All five overlapping cells agree.

3. Integrity `[measured]`

- **Selftest (10 checks)**, `audits/selftest.txt`:
  - the proved order equals a brute-force point count on 16 curves;
  - the batched sorted count equals an unbatched hash-map count on 30 bases of every kind, and across a chunk boundary;
  - the Z/N counter equals a naive pair-of-pairs count;
  - the integer row counts match hand values (0, 0, 2);
  - the base sizes equal ca5e01′s cells;
  - a whole job is a pure function of its labels.
- **Recount.** The unbatched recount of null, window, GAP and plant bases ran on every 97th curve of every rung, with no mismatch.
- **Determinism.** The 24-bit rung rerun with 2 workers instead of 6 is byte-identical (`audits/determinism.txt`).
- **Source.** The binary was built from the file hashed in the registration (`16d048d1…`), see `BINARY.sha256`. The raw per-curve outputs are 3.6 MB; their hashes are in `main/SHA256SUMS`, and `run.sh` regenerates them byte for byte.
- **Code.** Standard library only; rustfmt and clippy clean on the local toolchain.

4. What this closes and what it does not

- **Closed at scale.** For x-coordinate-structured bases (window, rank-2 and rank-3 GAP) at θ = 0.30 and 0.35, κ is constant in N over 16 to 32 bits, so the harvest law cannot reach break-even there: a constant κ at U = N^(1/(2h−1)) is N^(−1/(2h−1)) short (IDEA-20261018-a0830c).
- **Not closed.**
  - Growth beyond 32 bits (`p < 2^32`; larger rungs need 128-bit keys and, at θ = 0.35, memory for about 10^8 pair sums per base at 40 bits).
  - Non-geometric structure, and bb0d26's log-GAP label sets (a different statistic, on other curves).
  - Any family chosen from algebraic structure rather than coordinate geometry.
  - A reading from these rungs is a bound over 16 bits of N, not a law.
- **What reopens it.** A κ that is not flat across rungs for some structured family, or this ladder extended to 36–48 bits with wider keys.
- **Atlas note (not edited here).** `LEAD-ICP-coincidence-harvest-structured-base` names the bb0d26 independent review as its pending step; bb0d26's `analysis.md` (EV-PFDR-07d608, DEC-20261029-01405d) shows that review done. This study supplies the ladder step, so the lead's disposition can move to `exhausted_at_scale` on the evidence above.

5. Deviations and disclosures

- **R1: plant γ = 1/3 cannot be built** at θ = 0.30 or 0.35, because c_0 = N^(1/3) exceeds b/2 at every rung. The plants are γ = 0, 1/6 and 1/5 (PROTOCOL.md §6).
- **Dry run before registration** on a throwaway block (9,999,000) at 16–28 bits: the controls recovered and all real slopes were within ±0.016. The predictions were written after it (PROTOCOL.md §11).
- **No cost claim, no ranking.** κ is a yield statistic. The replication block was not run, because the reading is not ALIVE.
- **Formalization.** No theorem is stated, so no Lean file is owed.

6. Files

- `PROTOCOL.md`, `REGISTRATION.sha256`, `registered-at.txt`: the registration.
- `../../examples/coincidence_ladder.rs`: the native code (selftest, census, design, analyze, stage1).
- `run.sh`: pilot, design, main census, determinism rerun, analysis (a pure function of the labels).
- `pilot/`, `design.json`, `main/SHA256SUMS`, `analysis/analysis.json`, `analysis/stage1.json`, `audits/`, `stage1/ca5e01-cells.tsv`: inputs and outputs.
- `plot.mjs`, `tables.mjs`, `kappa_ladder.svg/png`, `slopes.svg/png`, `analysis/tables.md`: figures and tables (tooling).
- `report.pdf`: this README.

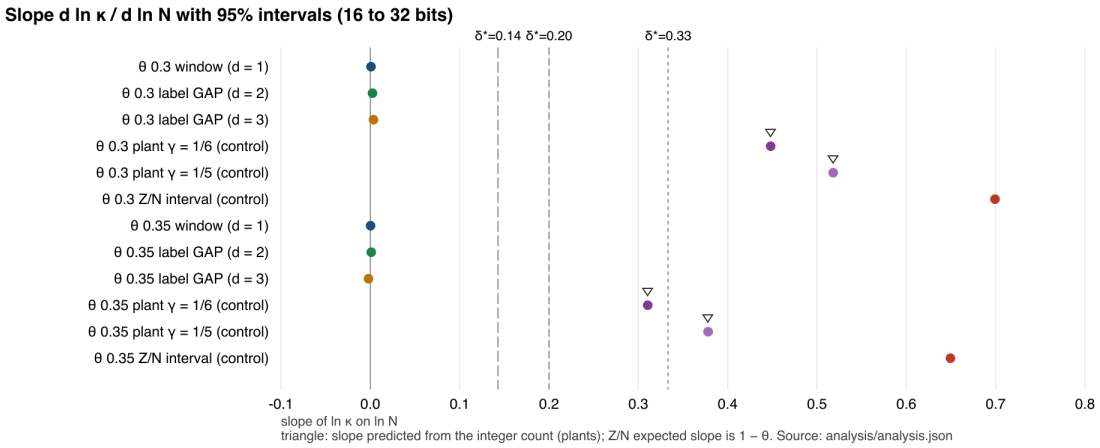


Figure 2: Slope of ln κ on ln N with 95% intervals for the six real combinations and the controls, against the break-even exponents. Triangles: slopes predicted from the integer count. Source: slopes.svg.